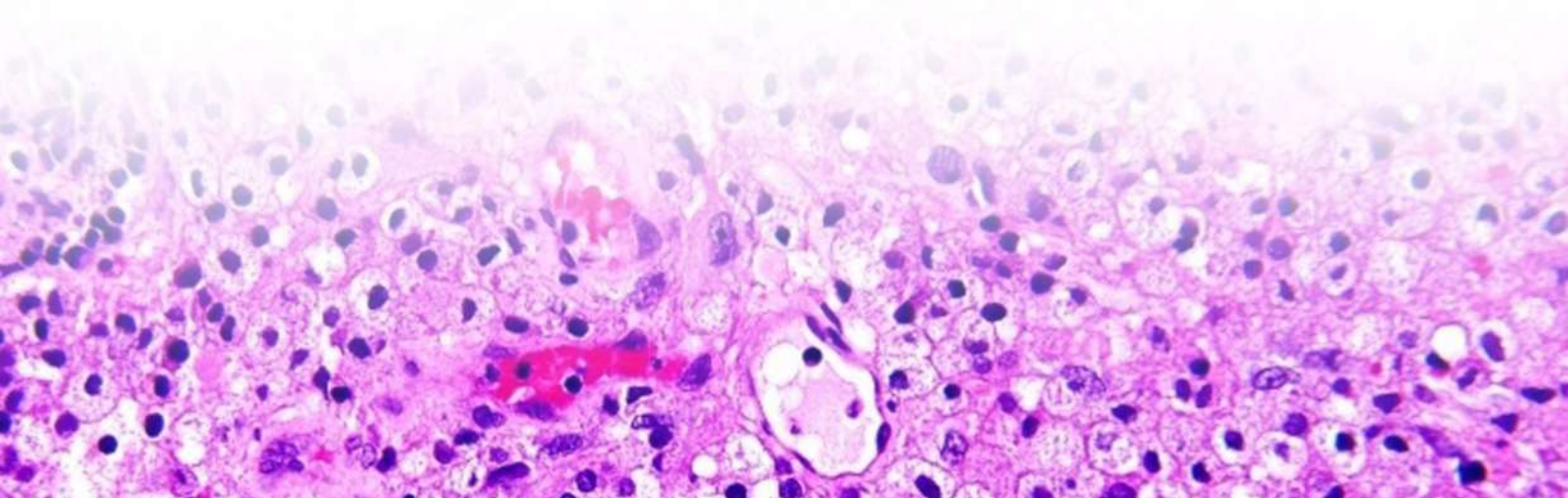
Student Notes

PATHOLOGY

1st Edition

Unit 19

Male Genital Tract



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

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LOWER URINARY TRACT AND MALE GENITAL SYSTEM

SHORT NOTES

1. Classical Seminoma
2. Cryptorchidism
3. Congenital anomalies of Urinary Bladder
4. Benign Prostatic Hyperplasia
5. Gynaecomastia
6. Bowen's Disease
7. Transitional Cell CA Bladder
8. Adenocarcinoma of Prostate

SHORT ANSWERS

1. Condyloma Accuminatum
2. Spermatocytic Seminoma
3. Difference between Classical and Spermatocytic Seminoma
4. Yolk Sac Tumours
5. Teratoma

UPDATES

LONG ANSWERS

1. SEMINOMA / CLASSICAL SEMINOMA / TYPICAL SEMINOMA:-

- Seminomas are the most common type of Germ cell tumor, making about 50% of Germ Cell Tumors.
- Seminomas almost never occur in infants.
- It resembles DYSGERMINOMA (ovarian tumor)
- Approximately 25% of these tumors have KIT activation mutation

MORPHOLOGY:

- **Gross:**
  - Seminomas produce Bulky masses, sometimes 10 times the size of normal testis.
  - The typical seminoma has a cut surface which is,
    - ✓ Homogenous
    - ✓ grey white
    - ✓ lobulated
    - ✓ Usually devoid of haemorrhage or necrosis.
- **Microscopically,**
  - It is composed of sheets of uniform cells divided into poorly demarcated lobules by delicate fibrous septa containing a lymphocytic infiltrate.
  - Seminoma cell is
    - ✓ Large
    - ✓ round to polyhedral
    - ✓ has a distinct cell membrane
    - ✓ clear or watery appearing cytoplasm
    - ✓ Large, central nucleus with 1 or 2 prominent nucleoli
    - ✓ The cytoplasm contains varying amount of Glycogen

- **Immunohistochemistry,**
  - Seminomas cells stain (+ve) for
    - ✓ KIT MUTATION
    - ✓ OCT4
    - ✓ placental alkaline phosphatase (PLAP)
  - Rare case may produce Beta-Human chorionic Gonadotropin (ß-HCG)
  - Approximately 15% of seminomas contain syncytiotrophoblast

2. CRYPTORCHIDISM:

- It refers to complete/partial failure of intra- abdominal testes to descend into scrotal sac.
- It is associated with testicular dysfunction & increased risk of testicular cancer.
- **Risk factors:**
  - Genetic disorders
  - Prematurity
  - Small for gestational age
  - Low birth weight (<2.5 kg)
  - In utero exposure to diethylstilbestrol / pesticides
- **Normal physiology of testicular descent:** Descent occurs in two phases
  - **First Phase: Transabdominal phase**
    - ✓ Testis comes to lie within the lower abdomen / brim of the pelvis.
    - ✓ This phase is controlled by the hormone called Mullerian – inhibiting substance.
  - **Second Phase: Inguino scrotal phase**
    - ✓ Testes descend through the inguinal canal into the scrotal sac.
    - ✓ This phase is androgen – dependent & is mediated by androgen – induced release of calcitonin gene – related peptide from genitofemoral nerve.
    - ✓ In cryptorchidism, the most common site of arrest is in the inguinal canal.

MORPHOLOGY:

- **Gross:**
  - Small & firm (due to fibrotic changes)
  - Unilateral in most cases
- **Microscopy:**
  - Histological changes in the malpositioned testis begin as early as two years of age which includes the following
  - Arrested germ cell development
  - Marked hyalinization & thickening of basement membrane of spermatic tubules thus tubules appears as dense cords of hyaline connective tissue outlined by a prominent basement membrane
  - Increase in interstitial stroma
  - Prominent Leydig cells as they are spared.
  - Similar histologic changes occur in C/L descended testis in males with unilateral cryptorchidism
  - Cryptorchidism is a marker of some intrinsic defect in testicular development, function and cellular differentiation that is unrelated to anatomic position

COMPLICATIONS OF UNDESCENDED TESTIS (CRYPTORCHIDISM)

- Sterility due to inhibition of spermatogenesis
- Prone to trauma & crushing injuries
- Inguinal hernia
- Testicular cancer

TREATMENT:

- During the first year, the majority of inguinal cryptorchid testes descend spontaneously into scrotum.
- If undescended, require surgical correction before 2 years of age that is before the onset of histologic deterioration.
- The procedure done is ORCHIOPEXY – placement of testes in scrotal sac.

3. CONGENITAL ANOMALIES OF URINARY BLADDER:-

- **Vesicoureteral reflux :-**
  - It is most common congenital anomaly, which is abnormal connections between the bladder and the vagina, rectum, or uterus.
- **Diverticula:-**
  - It is a pouch like evaginations of the bladder wall that is from less than 1 cm to 5 cm or 10 cm in diameter. May be acquired or congenital
  - These are mainly due to obstruction to urine out flow and marked thickening of the bladder wall.
  - The increased intravesical pressure causes outpouching of the bladder wall and formation of diverticula.
- **Exostrophy of the bladder:-**
  - It is a developmental failure in the anterior wall of abdomen and the bladder, so the bladder tend to communicate directly
  - As a result the bladder undergoes colonic glandular metaplasia and also the infections spread to the upper limit,which causes serious problems
- **Urachal anomalies:-**
  - The urachus (the canal that connects the fetal bladder with allantois) is normally obliterated but sometimes it may persist as a part or a whole.
  - Thus this makes a connection between the bladder and the umbilicus, thus making a threat.

4. BENIGN PROSTATIC HYPERPLASIA OR BPH:-

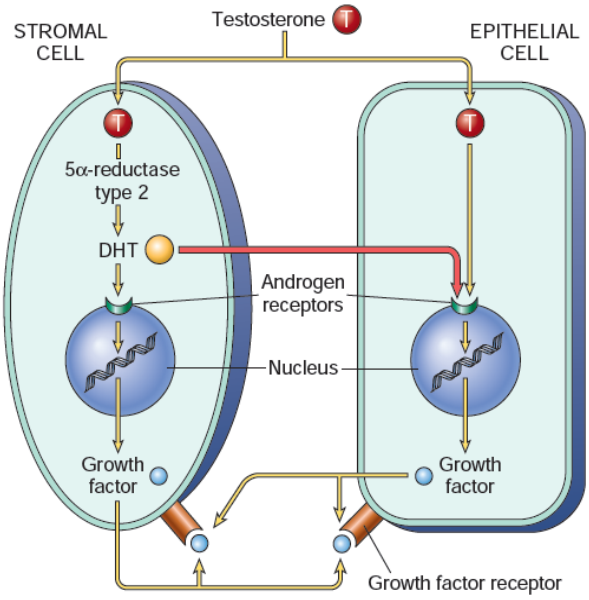
- It refers to the nodular (large, fairly, discrete nodules) hyperplasia of prostatic stromal & epithelial cells in the periurethral region of the prostate.
- It almost originates exclusively in the transition zone of the gland.
- Median lobe hypertrophy refers to the nodular enlargement of the prostate projecting into the floor of urethra as hemispheric mass directly beneath the mucosa of urethra.
- It often leads to urinary obstruction.
- Occurs > 50 yrs of age in males, Not a premalignant condition.

PATHOGENESIS:

- Hyperplasia is due to
  - Impaired cell death
  - Accumulation of senescent cells in the prostate Contributed by androgens.
  - Main androgen in the prostate is Dihydrotestosterone (about 90% of total prostatic androgens)
  - It is formed from testosterone by the action of an enzyme type 2, 5 alpha – reductase almost present entirely in stromal cells of prostate, absent in epithelial cells
- Thus, stromal cells responsible for androgen dependent prostate growth.
- DHT binds to its receptor on stromal and epithelial prostatic cells which stimulates the transcription of androgen dependent genes which includes several growth factors and their receptors.
- Some of the pathways which have produced embryonic prosthetic development maybe "reawakened" in adulthood to produced prostatic growth in BPH

MOA OF DIHYDROTESTOSTERONE:

- Important growth factors are:
  - **Fibroblast growth factor** – produced by stroma cells are paracrine stimulators of androgen stimulated epithelial growth during embryonic prostatic development.
  - **Transforming growth factor beta** – mitogen for fibroblasts & other mesenchymal cells but inhibits epithelial proliferation.
- Finally, DTH induced growth factors act by
  - ✓ Increasing proliferation of stromal cells
  - ✓ Decreasing death of epithelial cells



MORPHOLOGY:

- **Gross:**
  - Enlarged prostate of 60-100 gm weight
  - Contains well – circumscribed nodules that bulge from cut – surface.
  - Early nodules-composed almost entirely of stromal cells
  - Later predominantly epithelial nodules arises
- **On cross section:**
  - Nodules vary in colour & consistency depending on their cellular content.
  - **Nodules containing glands** – yellow pink & soft, Exudes milky white prostatic fluid.
  - **Nodules containing fibromuscular stroma**-pale grey and tough, Do not exudes fluid
  - True capsule is absent
- **Microscopy :**
  - Glandular proliferation– aggregation of small to large cystically dilated glands
  - Lined by two types of cells: Inner columnar & Outer cuboidal / flattened epithelium
  - Glandular lumina contain laminated proteinaceous secretory substance known as corpora amylacea
  - In prominent BPH, foci of reactive squamous metaplasia mimicking urothelial carcinoma are seen adjacent to prostatic infarcts.

CLINICAL FEATURES

The major clinical problem is urinary obstruction. Features of urinary obstruction such as

- Increased urinary frequency, urgency, hesitancy
- Bladder hypertrophy, destination and urine retention
- Difficulty in starting & stopping the stream of urine
- Increased risks of bacterial infections of bladder & kidney
- Overflow dribbling
- Dysuria
- Nocturia
- Acute urinary retention require emergency catheterization for relief

TREATMENT:

- Mild cases of BP –Decreasing fluid intake especially before bedtime, moderating the intake of alcohol and caffeine containing products, and time voiding schedules
- Symptomatic treatment using alpha 1 adrenergic blockers (decrease the prostatic smooth muscle tone),5 alpha reductase inhibitors (cause shrinkage of prostate gland by decreasing the synthesis of DHT).
- Surgical treatment **– TURP (Transurethral resection of prostate)**-gold standard treatment, high intensity focused ultrasound, laser therapy, hyperthermia, transurethral needle ablation using radio frequency e and trans urethral electro vaporisation.

5. GYNAECOMASTIA:

- It is defined as enlargement of male breast, a benign one.
- It occurs as a button like subareolar enlargement.
- May be unilateral or bilateral.
- May occur during puberty or very aged or during adult life etc.
- **Pathogenesis:** It occurs as a result of imbalance between oestrogen (breast tissue producing hormone) and testosterone (counteracts the action of oestrogen).
- **Causes:-**
  - Cirrhosis of liver, since this organ is responsible for metabolizing of oestrogen
  - Increase in oestrogen level, as this causes suppression of testosterone.
  - Or due to alcohol, marijuana, heroin, antiretroviral therapy and anabolic steroids.
  - Rarely may be part of klinefelter's syndrome.
  - Or a part of testicular tumours such as leydig or sertoli cells.
- **Histology:** There is an increase in dense collagenous connective tissue with epithelial hyperplasia of the duct lining the micropapillae

6. BOWENS DISEASE:

- There are two distinct lesions display histologic features of CIS (Carcinoma In situ) in external male genitalia :
  - ✓ Bowen disease
  - ✓ Bowenoid papulosis
- These lesions have a strong association with infection by high – risk HPV, most commonly type 16.

BOWENS DISEASE:

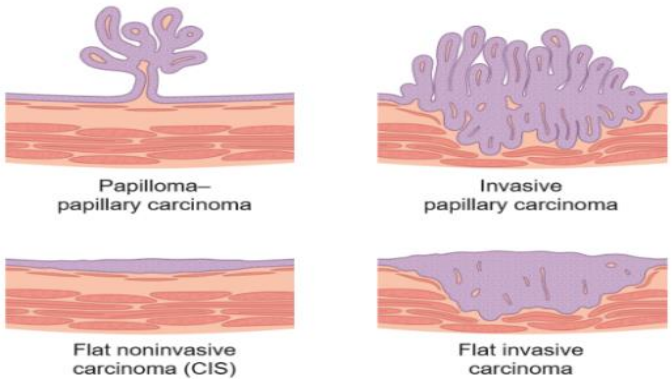
- It occurs in the genital region of both men & women older than 35 years.
- In men, it tends to involve the skin of the shaft of the penis & scrotum.
- **Gross:**
  - Solitary, thickened, gray-white plaque with ulceration & crusting
  - Rarely as red velvety plaque – Erythroplasia of Queyrat
- **Histology:**
  - Hyperproliferativeepidermis with atypical mitoses
  - Dysplastic cells with hyper chromatic nuclei
  - Lack of orderly maturation.
  - Dermal – epidermal border is sharply delineated by an intact basement membrane.
  - 10% transforms into malignancies (squamous cell carcinoma

BOWENOID PAPULOSIS: Differs from Bowen disease by

- Younger age of onset especially sexually active adults.
- Multiple (rather than solitary) reddish brown pigmented papular lesions.
- Never develops into invasive carcinoma.
- Regress spontaneously.
- Histologically indistinguishable from bowen disease

7. TRANSITIONAL / UROTHELIAL CELL CARCINOMA BLADDER:-

- About 95% of bladder tumors are Epithelial Origin
- Most epithelial tumors are urothelial (transitional cell) type and are thus interchangeably called urothelial or transitional tumor
- There are two distinct precursor lesions to invasive urothelial carcinoma:
  - **Noninvasive papillary tumors:** Most Common, Originate from papillary urothelial hyperplasia
  - **Flat noninvasive urothelial carcinoma:** Also known as Carcinoma In Situ (CIS)
- **Epidemiology & pathogenesis:**
  - Incidence of carcinoma of Bladder is higher in male than in female
  - Almost 80% of patients are between 50-80 years of age
- **Risk factors**
  - Phenacetin (Mnemonic: **P**ee **SAC**)
  - Smoking
  - Aniline: compound used in rubber/dye manufacturing
  - Cyclophosphamide



- **Genetic alterations**
  - Gain of Mutation in FGFR3
  - HRAS mutation
  - Loss of Mutation in TP53
  - Loss of Mutation in RB
  - Particularly common are losses of genetic material on chromosome 9

MORPHOLOGY:

- The appearance of urothelial tumors varies from purely papillary to nodular or flat
- Most arise from the lateral or posterior walls at the bladder base

- **A. Papillomas**
  - Represent 1% or less of bladder tumours
  - Seen in younger patients
  - These tumors typically arise singly as small, delicate structure, superficially attached to mucosa by a stalk & are referred to as EXOPHYTIC PAPILLOMA
  - The individual finger like papillae have central core of loose fibrovascular tissue covered by a epithelium that is histologically identical to normal urothelium
- **B. Papillary urothelial neoplasm of low malignant potential**
  - Share many histologic features with papilloma, differing only in having thicker urothelium
  - Progression to tumors of highest grade may occur but rare
- **C. Low grade papillary urothelial carcinoma**
  - Orderly architectural &cytologic appearance
  - Cells are evenly spaced & cohesive
- **D. High grade papillary urothelial carcinoma**
  - Contain discohesive cells with large hyperchromatic nuclei
  - Some of tumor cells are highly anaplastic
  - Mitotic figures including atypical ones are present
  - Loss of polarity is seen
  - 80% of these type are invasive
- **E. Carcinoma in situ (CIS)**
  - Flat urothelial carcinoma
  - Presence of cytologically malignant cells within flat urothelium
  - Lack of cohesiveness, leads to shedding of malignant cells in urine
  - CIS may range from full-thickness cytologic atypia to scattered malignant cells in an otherwise normal urothelium, the latter termed as PAGETOID SPREAD
  - CIS usually appears as an area of mucosal reddening, granularity, or thickening without an evident intraluminal mass
  - It is commonly multifocal and may involve most of the bladder surface
  - If untreated, 50% to 75% of CIS cases progress to invasive cancer
- **F. Invasive urothelial cancer**
  - May be associated with papillary urothelial cancer, usually high grade, or adjacent CIS.

CLINICAL COURSE:

- Bladder tumors classically produce painless hematuria
- Hematuria accompanied by Frequency, Urgency, Dysuria
- When the ureteral orifice is involved, pyelonephritis or hydronephrosis may follow
- Individuals with urothelial tumors, whatever their grade, have a tendency to develop new tumors after excision, and recurrences may show a higher grade
- Although the term recurrence is used, most of the subsequent tumors arise at different sites from the original lesion

DIAGNOSIS:

- The clinical challenge with these neoplasms is early detection and adequate follow up
- For tumors detected at an earlier stage, cystoscopy and biopsy are the mainstays of diagnosis
- Additionally, cytologic examination of cells obtained from urine samples and tests performed on urine to detect chromosomal abnormalities (aneuploidy of chromosome 3, 7, and 17 and 9p deletions) by fluorescent in situ hybridization (FISH) are also helpful screening measures, particularly for CIS that might be missed by cystoscopy

TREATMENT:

- For small, localized low-grade papillary tumors, the diagnostic transurethral resection is the only surgical procedure done
- Patients at high risk of recurrence and/or progression like
    - CIS
    - Papillary tumors that are
        - ✓ High grade
        - ✓ Multifocal
        - ✓ Have a history of recurrence
        - ✓ Associated with lamina propria invasion

  receive intravesical instillation of an attenuated strain of Mycobacterium bovis called bacillus Calmette-Guérin (BCG). The bacteria elicit a local inflammatory reaction that destroys the tumor.
- Radical cystectomy is typically reserved for
    - tumor invading the muscularispropria
    - CIS or high-grade papillary cancer refractory to BCG
    - CIS extending into the prostatic urethra and into the prostatic ducts, where BCG will not come into contact with the neoplastic cells.

8. ADENOCARCINOMA OF PROSTATE:-

- Adenocarcinoma of the prostate is the most common form of cancer (29%) accounting men in the United States.
- **Incidence:-**
  - It affects men older than the age 50 yrs.
  - It is most common among blacks.
  - It may be due to switch to modern food habits and lifestyle changes.

PATHOGENESIS:-

- Androgens play an important role in prostate cancer. The androgen receptor gene contains repeated sequences of codon CAG.
  - The androgen receptor is the one which prostate cancer cells tend to join.
  - If the repeat sequence is shortest, the incidence is more.
- Men with germline mutation in BRCA2 and in HOXB13 have increased risk of prostate cancer.
- The common genetic change in prostate cancer is Chromosomal rearrangements that juxtapose the coding sequence of ETS family transcription factor gene (ERG or ETV1) next to androgen regulated TMPRSS2 promoter.
- Amplification of 8q24 locus containing the MYC oncogene and deletions involving PTEN tumor suppressor.
- Loss of TP53 and deletions involving RB are common.
- There occurs hypermethylation of glutathione s - transferase (GSTP1) gene which downregulates GSTP1 expression. The GSTP1 gene is as crucial for preventing carcinogens.

MORPHOLOGY:-

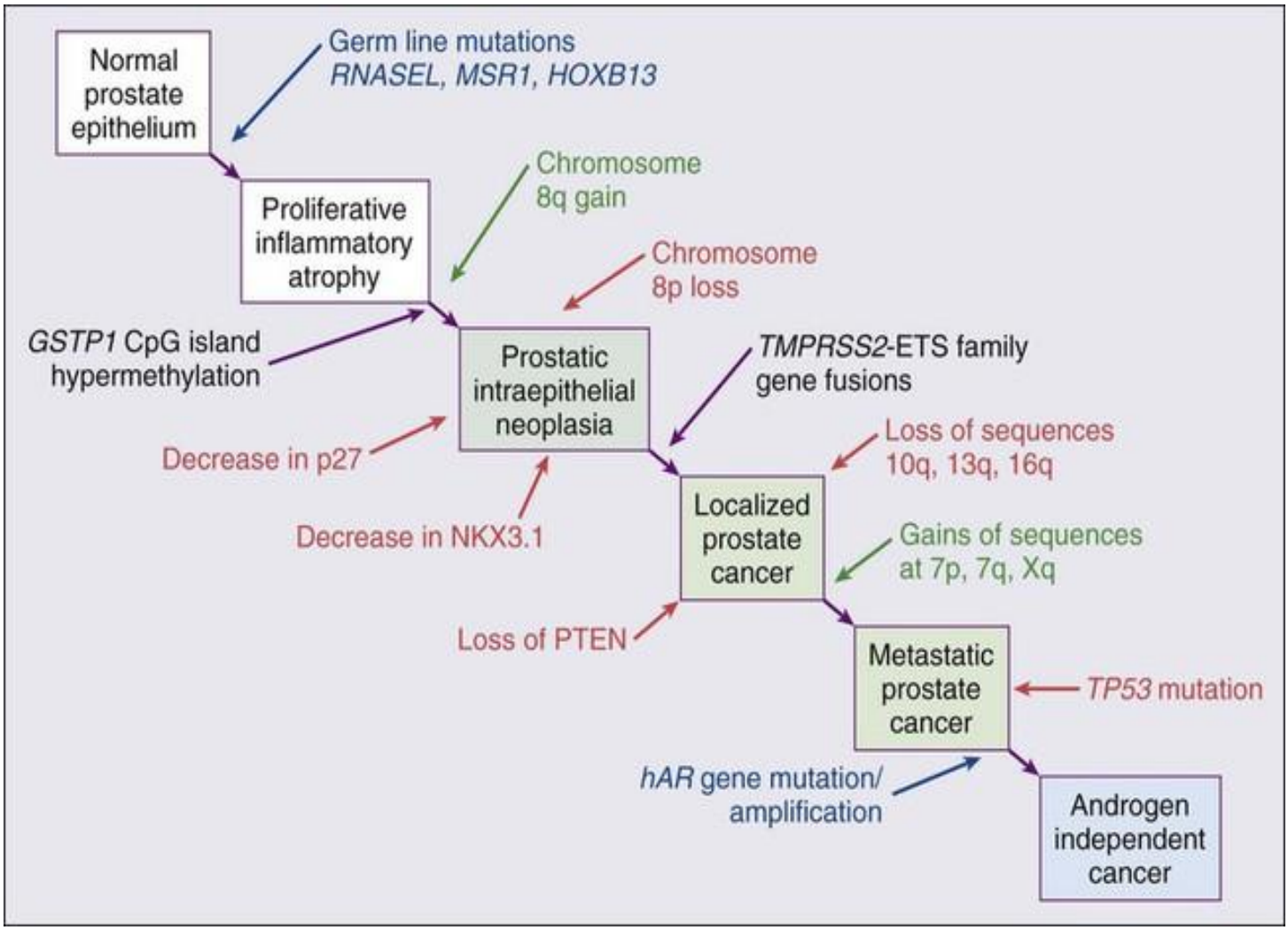
- **Gross :-**
  - Carcinoma of the prostate arises in the peripheral zone of the gland, a posterior area that is palpable on rectal examination.
  - It mostly involves periprostatic tissue, seminal vesicles, and the base of the urinary bladder.
- **Histology:-**
  - It contains glands lined by a uniform layer of cuboidal or low columnar epithelium.
  - These glands are more crowded but lack branching and papillary infolding.
  - The cytoplasm is pale - clear to amphophilic appearance.
  - The nuclei is large and contains one or more nucleoli.

Precursor lesion: high grade prostatic intraepithelialneoplasia (Pin):-

- It is a putative precursor lesion for prostatic carcinoma.
- PIN progresses to invasive cancer mostly.
- Higher frequency & greater extent of PIN seen in proximity to the site of cancer.

MOLECULAR CHANGES IN PIN:

- Chromosomal Rearrangements that juxtapose the coding sequence of an ETS family transcription factor gene (most commonly ERG / ETV1) next to androgen regulated TMPRSS2 promoter.
- Leads to over expression of ETS transcription factors → invasiveness of prostate epithelial cells → upregulation of matrix metalloproteinases.

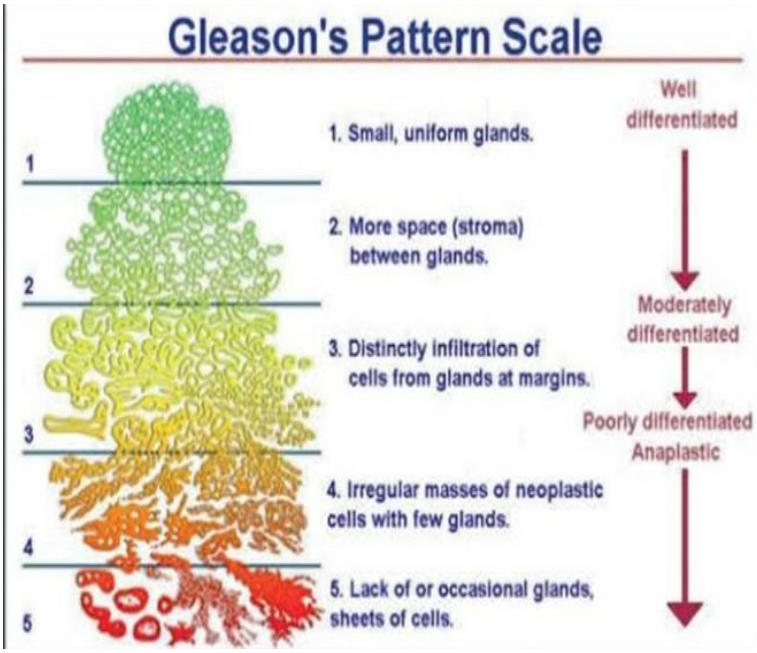


MORPHOLOGY IN HIGH GRADE PIN:

- In approximately 80 percentage of cases, prostatic tissue removed also harbours presumptive precursor lesions refer to as high grade prostatic intraepithelial neoplasia
- Benign , large, branching prostatic acini
- Lined by cytologically atypical cells with prominent nuclei
- Glands are surrounded by a patchy layer of basal cells & an intact basement membrane.

Gleason grading of prostate cancer: otb of level of glandular differentiation:

New category	Histological descriptions of new grading categories
Grade 1	• Only individual, discrete well-formed glands
Grade 2	• Predominantly well-formed glands with lesser component of poorly formed, fused, and/or cribriform glands
Grade 3	• Predominantly poorly formed poorly formed, fused, and/or cribriform glands with lesser component of well-formed glands
Grade 4	• ***Only*** poorly formed, fused, and/or cribriform glands ***or*** • Predominantly well-formed glands with lesser component lacking gland formation ***or*** • Predominantly lacking gland formation and lesser component of well-formed glands (but poorly formed, fused, and/or cribriform glands can be a minor component)
Grade 5	• Lacks gland formation – or has glands with necrosis -- with or without poorly formed, fused, and/or cribriform glands



SHORT ANSWERS

1. CONDYLOMA ACCUMINATUM:-

- It is a benign sexually transmitted wart caused by human papillomavirus (HPV) - commonly type 6 and 11.
- It occurs in moist muco-cutaneous surfaces of the external genitals of either sex.
- Can recur but rarely progresses to tumor.
- **Gross:** These occur in single or multiple sessile or pedunculated red papillary excrescences.
- **Histology:** A branching, villous papillary connective tissue stroma is covered by epithelium with acanthosis (superficial keratosis and thickening of underlying epidermis)
- **Kilocytosis:** cytoplasmic vacuolization of squamous cells, characteristic feature of HPV

2. SPERMATOCYTIC SEMINOMA:-

- It is a rare, not similar to seminoma, slow growing germ cell tumour.
- It affects predominantly older men aged 65 and above
- **Gross:** It consists of soft,pale grey, cut surfaces that sometimes reveal mucoid cysts.
- **Histology :**
  - It contains three cell populations intermixed medium sized cells (containing round nucleus and cytoplasm), smaller cells, and scattered giant cells.
  - But compared to classical seminomas, these lack lymphocytes, granulocytes, syncytiotrophoblasts, extra testicular sites of origin etc.

3. DIFFERENCE BETWEEN CLASSICAL AND SPERMATOCYTIC SEMINOMA:-

Classical seminoma	Spermatic seminoma
Most common type of Germ cell tumor	Rare type of Germ cell tumor
Originate from Intratubular germ cell neoplasia (ITGCN)	Uncertain Origin (not from ITGCN)
Predominantly affects infants & children	Predominantly affects older men
Lymphocytic infiltrate is present	Lack lymphocytes
Granulomas seen with tuberculosis	Lack granulomas
Syncytiotrophoblast rarely seen	Lack Syncytiotrophoblast

4. YOLK SAC TUMOR / ENDODERMAL SINUS TUMOR:-

- Most common testicular tumor in Infants & Children upto 3 Years.
- In adults the pure form of this tumor is rare; instead yolk sac elements frequently occur in combination with embryonal carcinoma
- **Morphology:**
  - These tumors are Non-encapsulated & have a homogenous, yellow white, mucinous appearance.
  - They are composed of a lacelike (reticular) network of medium sized cuboidal or flattened cells.
  - In approximately 50% of tumors, structures resembling endodermal sinuses (**schiller-duval bodies**) may be seen; it resembles primitive glomeruli
- **Immunohistochemistry:** Present within or outside the cytoplasm are eosinophilic, hyaline like globules in which **Alpha Fetoprotein** and **Alpha-1 antitrypsin** can be demonstrated.

5. TERATOMA:-

- It is second most common among infants and children.
- It contains organoid components reminiscent of the normal derivatives of more than one germ layer.
- It may progress to malignancy.
- **Gross:** It is usually large, ranging from 5 to 10 cm in diameter. The structure is solid or cartilage nous, and cystic areas, Hemorrhage and necrosis is seen.
- **Histology:-** It contains neural tissue, muscle bundles, islands of cartilage, clusters of squamous epithelium, structures reminiscent of thyroid gland, bronchial or bronchiolar epithelium, and bits of intestinal wall or brain tissue.
- **Immunohistochemistry:** These stain positive for isochromosome 12p.

UPDATES

1. CYSTITIS

- Emphysematous cystitis- C. perfringens associated gas filled vesicles in bladder wall.
- Iatrogenic cystitis- associated with pelvic irradiation

2. TRANSITIONAL CELL TUMOR / UROTHELIAL NEOPLASMS:

- Noninvasive urothelial (transitional cell) tumor
- Infiltrating urothelial carcinoma
- Variants: nested, microcystic, micropapillary, plasmacytoid, sarcomatoid, giant cell, poorly differentiated, lipid-rich, and clear cell.

Precursor lesion	Mutation associated	Cancer type
Urothelial Papilloma	RAS activation , amplification of FGFR3	Non-muscle invasive papillary Ca (most common) (or) Muscle Invasive papillary Ca (20%)
Flat noninvasive Urothelial CIS	TP53, RB Lof mut.	Flat invasive Ca

3. PENILE CANCER

- Invasive SCC develops from CIS lesions- Penile Intraepithelial Neoplasia (PeIN)
- PeIN may be: HPV associated (or) Non HPV associated

HPV associated PeIN	Non HPV related PeIN
• Undifferentiated	• Differentiated
• Minor number	• Most common
• Bowen's disease -older people Bowenoid papulosis- young	• Poor hygiene, Balanitis xerotica obliterans (lichen sclerosis et atropica) in older

- Non HPV related SCC of penis is called Conventional or differentiated type.
- Morphology: Macroscopically the tumors may be irregular, fungating cauliflower-like masses; flat, indurated lesions; or large verruciform/papillary tumors .
- Pathologic prognostic factors in penile carcinoma include stage, grade, histologic subtype, vascular invasion, and perineural invasion.
- Verrucous and papillary carcinomas are well differentiated/grade 1 tumors, while sarco-matoid and basaloid carcinomas are poorly differentiated/grade 3 tumors.
- Basaloid carcinoma is an HPV-related tumor comprised of relatively small hyperchromatic cells that has a destructive pattern of invasion and usually pursues an aggressive course.

4. ITGCN

- All testicular neoplasms have a common precursor- Intratubular Germ Cell Neoplasm (ITGCN) Or Germ Cell Neoplasia Insitu, except:
    - Pre-pubertal teratoma
    - Pre-pubertal Yolk sac tumor
    - Spermatocytic seminoma
- ITGCN is present in undescended testis.
- ITGCN is present even in utero. Stays dormant till puberty when hormones start ↑
- 70% cases of ITGCN develop Testicular Ca within 7 years. Thus ITGCN is mostly PROGRESSIVE precursor lesion.
- Activating mutations on KIT & isochromosome 12p are seen in majority of cases of ITGCN.

5. TESTICULAR NEOPLASMS

- Seminomas are positive for PODOPLANIN.
- Spermatocytic seminoma has gain of 9q characteristically.
- Pre- pubertal teratomas are benign; post- pubertal are malignant.
- Pathologic features correlated with malignancy in sex cord stromal tumors of the testis :
    - large size (greater than 5 cm),
    - necrosis,
    - infiltrative borders,
    - anaplasia,
    - mitotic activity,
    - vascular-lymphatic invasion, and
    - Extra testicular extension.
- Leydig cell tumor is associated with
    - Klinefelter syndrome,
    - Cryptorchidism,
    - Hereditary leiomyomatosis and
    - Renal cell carcinoma syndrome (caused by mutations in the fumarate hydratase)
- Sertoli cell tumor: associated with
    - Carney complex (caused by germline mutations in the gene PRKAR1A, which encodes a cyclic adenosine monophosphate–dependent protein kinase),
    - Peutz-Jeghers syndrome and
    - Familial adenomatosis polyposis syndrome
- Testicular lymphomas are frequently bilateral and involve the spermatic cord.

6. PROSTATIC CARCINOMA

- Polymorphisms in glutathione-S-transferase (GSTP1), an enzyme involved in detoxification of polycyclic aromatic hydrocarbons, are linked to prostate cancer risk.
- Amplification of MYC and deletion of PTEN- increase resistance to anti androgen therapy.
- The most aggressive variant of prostate cancer is small-cell carcinoma

7. ONE LINERS

- Multifocal keratinising squamous dysplasia of bladder → Dysplasia / Ca in situ → invasive squamous cell Ca bladder
- Estrogens contribute to BPH pathogenesis by tipping the balance toward proliferation. (ER alpha – increases proliferation; ER beta - decreases proliferation)
- **Peyronie disease**: Accumulation of collagenous plaques between Corpora cavernosa and Tunica albuginea. The fibrosis is the product of microvascular trauma and subsequent organizing sclerosing chronic inflammation.

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